

Please check that this question paper contains 9 questions and 2 printed pages within first ten minutes.

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Uni. Roll No.

12 DEC 2023

Program: B.Tech. IT (Batch 2018 onward)

Semester: 3rd

Name of Subject: Object Oriented Programming using C++

Subject Code: PCIT-102

Paper ID: 16041

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) What do you mean by a token?
- b) Define unary operator. Compare it with if and if-else statement.
- c) Given a C++ code snippet:

```
int main()
{ int num1 = 20;
  int num2 = 0;
  int result = num1 / num2;
  std::cout<< "Result: " << result << std::endl;
  return 0; }
```

Identify and explain any syntax errors or logical errors that you will find in the code

- d) Write a program to find the largest of three numbers using ternary operator.
- e) List some practical applications or scenarios where declaring and initializing arrays in C++ would be a beneficial programming approach
- f) What is the need of Object Oriented Programming paradigm?

Part – B

[Marks: 04 each]

- Q2.** Explain call by reference and call by value with example. Write a Program to swap numbers in cyclic order using call by reference .

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- Q3. List out visibility of inherited members in various categories of inheritance.?
- Q4. Elaborate the principles of structured and object-oriented development. How they impact the efficiency and scalability of software projects in different contexts.
- Q5. List out various situations for execution of base class constructor in inheritance.
- Q6. Examine the significance of member functions within a class. Illustrate your analysis by providing an example in which you thoroughly assess the role of the class's member functions and how they collectively enhance the class's overall functionality.
- Q7. Compare and contrast the variables and constants in C++. What are the rules to be followed for identifiers?

[Marks: 12 each]

Part – C

- Q8. Design a menu-driven program that encompasses all functionalities: Sum of Digits and odd number checking. (Make use of concepts like user-defined functions for the stated two tasks, parameter passing)

OR

Differentiate between break and continue statement. Why the use of goto statement is not good for quality programming?

- Q9. Design a C++ program that models a basic arithmetic calculator with support for both integer and floating-point numbers. Implement unary and binary operator overloading to enable the calculator to perform operations like addition, subtraction, multiplication, and division on user-provided operands.

OR

Develop a C++ program to create a simple application for managing different types of food. Define a base class "Food" with attributes like name, cuisine, and a virtual function "displayInfo" that displays the basic information about the food. Then, derive two classes, "Dosa" and "Dessert," from the base class, each with their own unique attributes and override the "displayInfo" function in each derived class to provide specific information about the food type. In your program's main function, create objects of both the "Dosa" and "Dessert" classes, and use a loop to display the information of each food using the base class pointer.
